Thread - 2024-12-01

https://x.com/RiikonenMarko/status/1863187134204490169

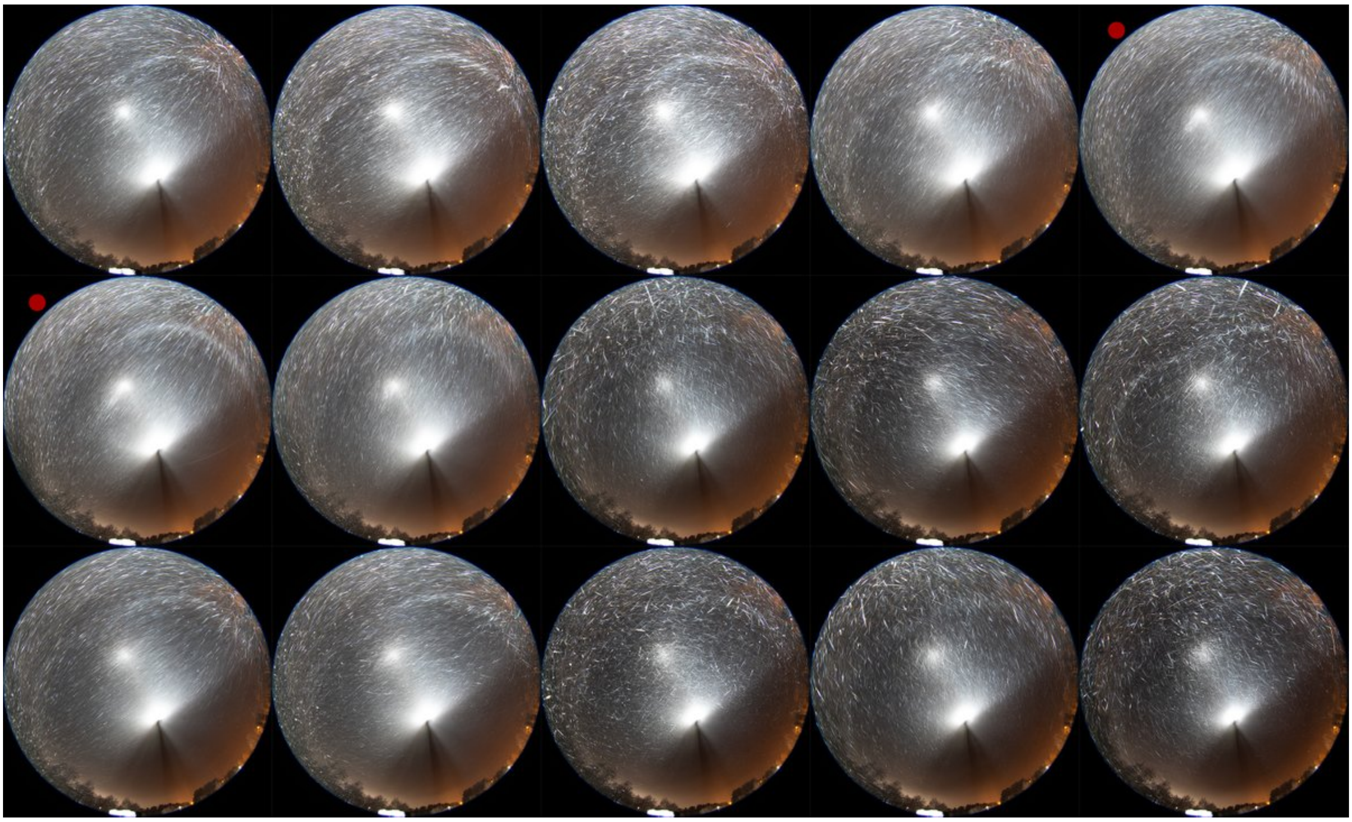
1/9 - 2024-12-01 11:43 UTC

There was a third successive night of ice fog (28/29 November 2024 Rovaniemi). Normal fog turned into ice crystals in -11 to -9 C temps, comfortably in the range of massive displays. However, the fog layer was too thick. Only when it thinned so as for the snow gunning to punch through it and make patches of starry sky appear, the quality stuff appeared. And even then only momentarily. Here is the last show of the night before the ice fog disappeared. I opted this time for the very low lamp.

Seems like wind has disrupted the crystals orientations in some of those shots, resulting in one sided counterhelic (subhelic) arc. In this collage of successive 30s exposures with a fixed camera-lamp setup, the effect is particularly stark in photos 5 and 6 (red dots). These two frames had the highest wind as indicated by the most streamlined crystals. Judging by the general crystal pattern the wind was coming from 1:30 direction. The disappeared counterhelic arc half coinsides with countersun (supersun) tail pointing in the direction opposite to the wind.

The crystal orientation disruption happens when wind meets an obstacle, here the camera. One expects turbulence on the lee side and randomization of crystal orientations. That would seem to explain the disappeared counterhelic arc half. But not of course the 7:30 direction pointing countersun tail, which is from slightly tilted crystals. Countersun from column orientation is weak and it must have been here chiefly made by plate oriented crystals. The obstacle aerodynamics resulted in a different effect for this halo.

Years ago I photographed a spotlight display where momentarily the other one of the counterparhelia (superparhelia) goes missing in a series of photos. Unfortunately these photos got accidentally deleted.



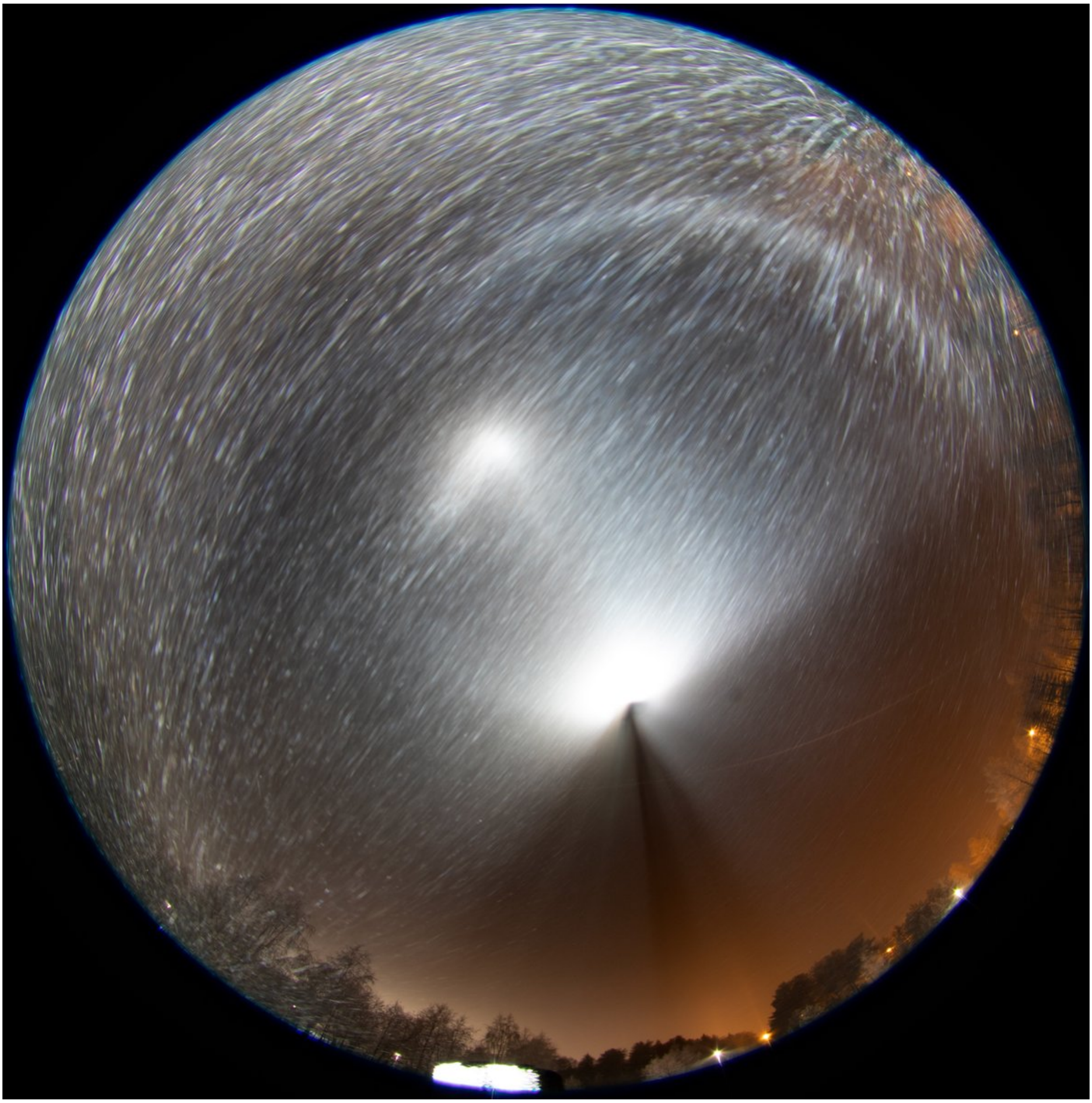
2/9 - 2024-12-01 12:21 UTC

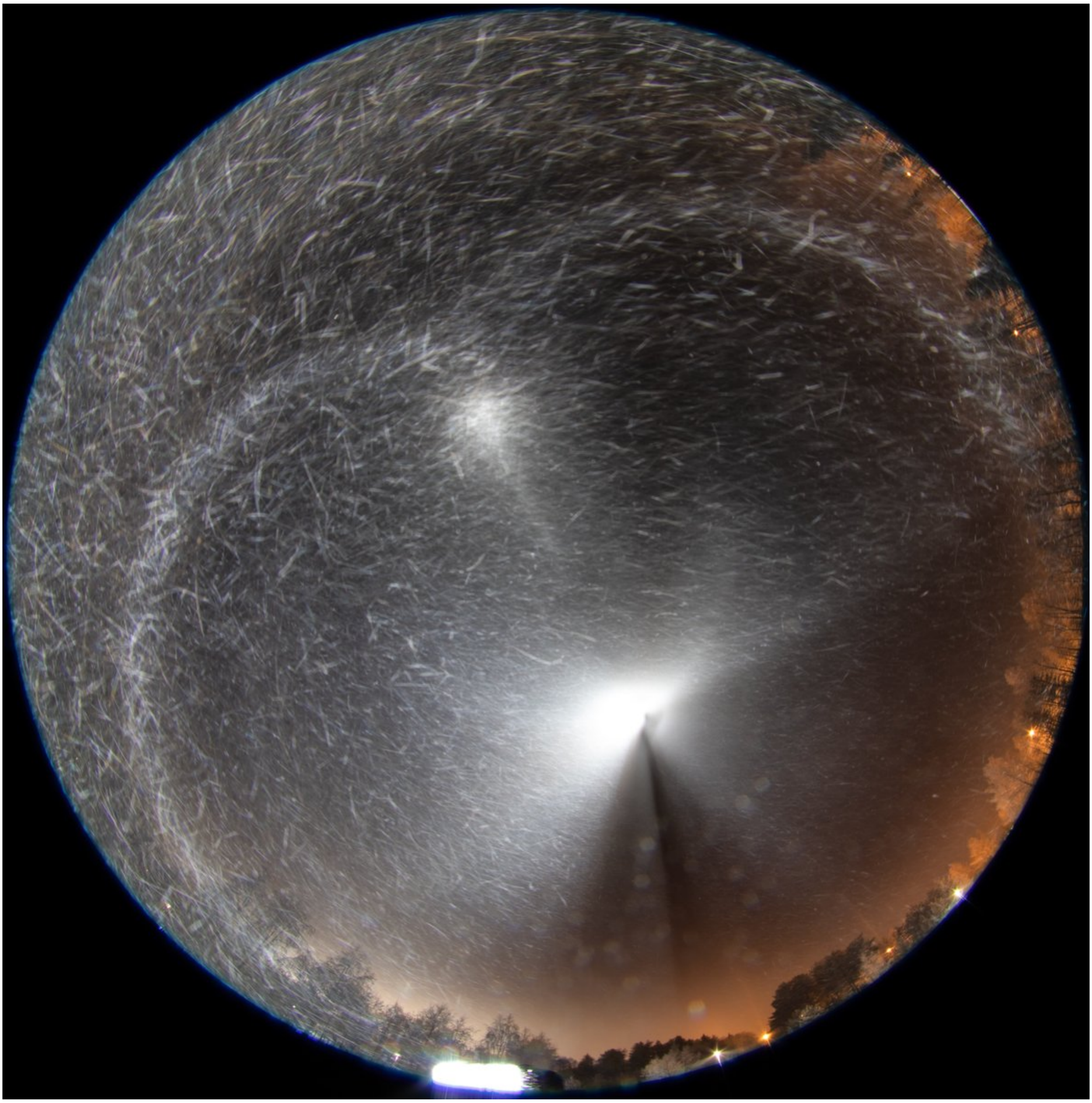
Some more images. The first one is a stack of most frames in the previous post collage. The second image has frames 5 and 6 stacked The third one has the lamp at a little less steep elevation. The intensity difference in the two sides of the counterhelic arc here look to be due to the beam not exactly having centered on the camera – the other side of the image is darker overall.

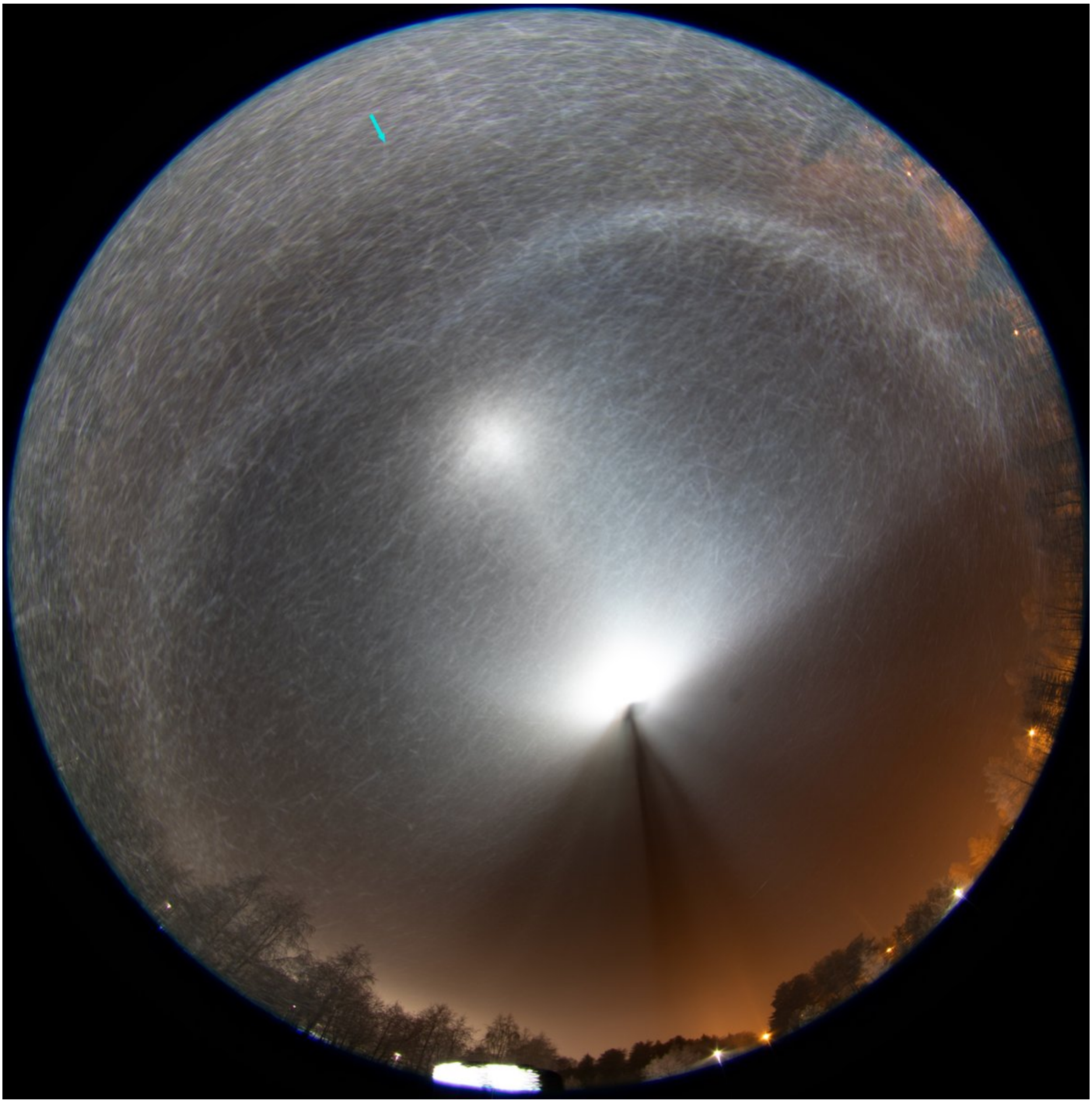
And then a photo showing the setup. At such short distances the area around the camera in which the display forms is really tight. This Cyclops lamp came with a halogen bulb, but we replaced it with hid for more brightness and less electricity consumption. As it has aged it has developed a personality. There is no more guarantee it starts. If it doesn't, I leave it attached to the battery with the switch on and use either one of my led two lamps. At some point then it will start just by itself and shows these leds who's the boss.

The arrow in the first image points to column 135, which is kind of like a weak copy of the infralatarc on the other side of the horizon.

28/29 November 2024 Rovaniemi



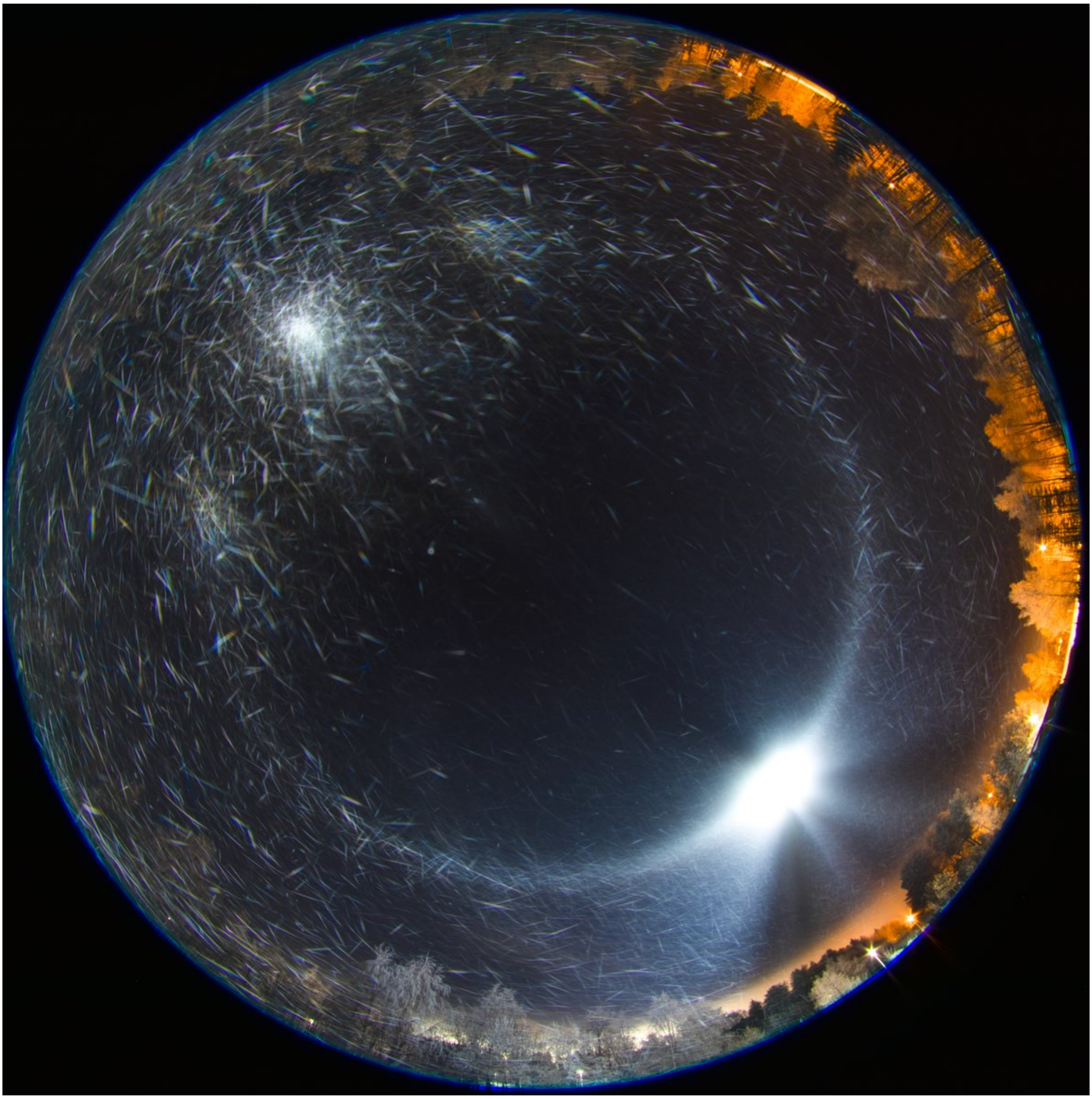


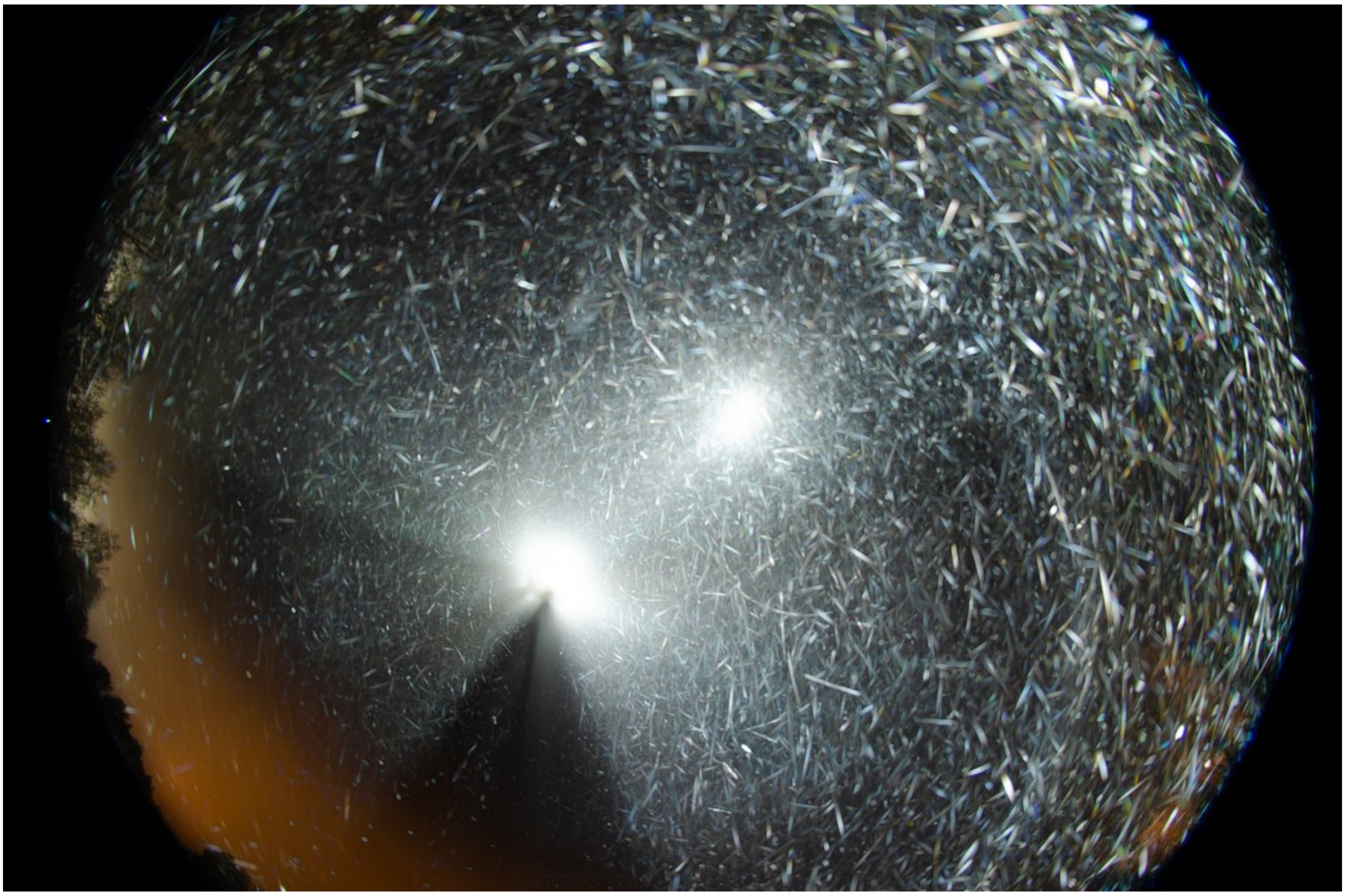


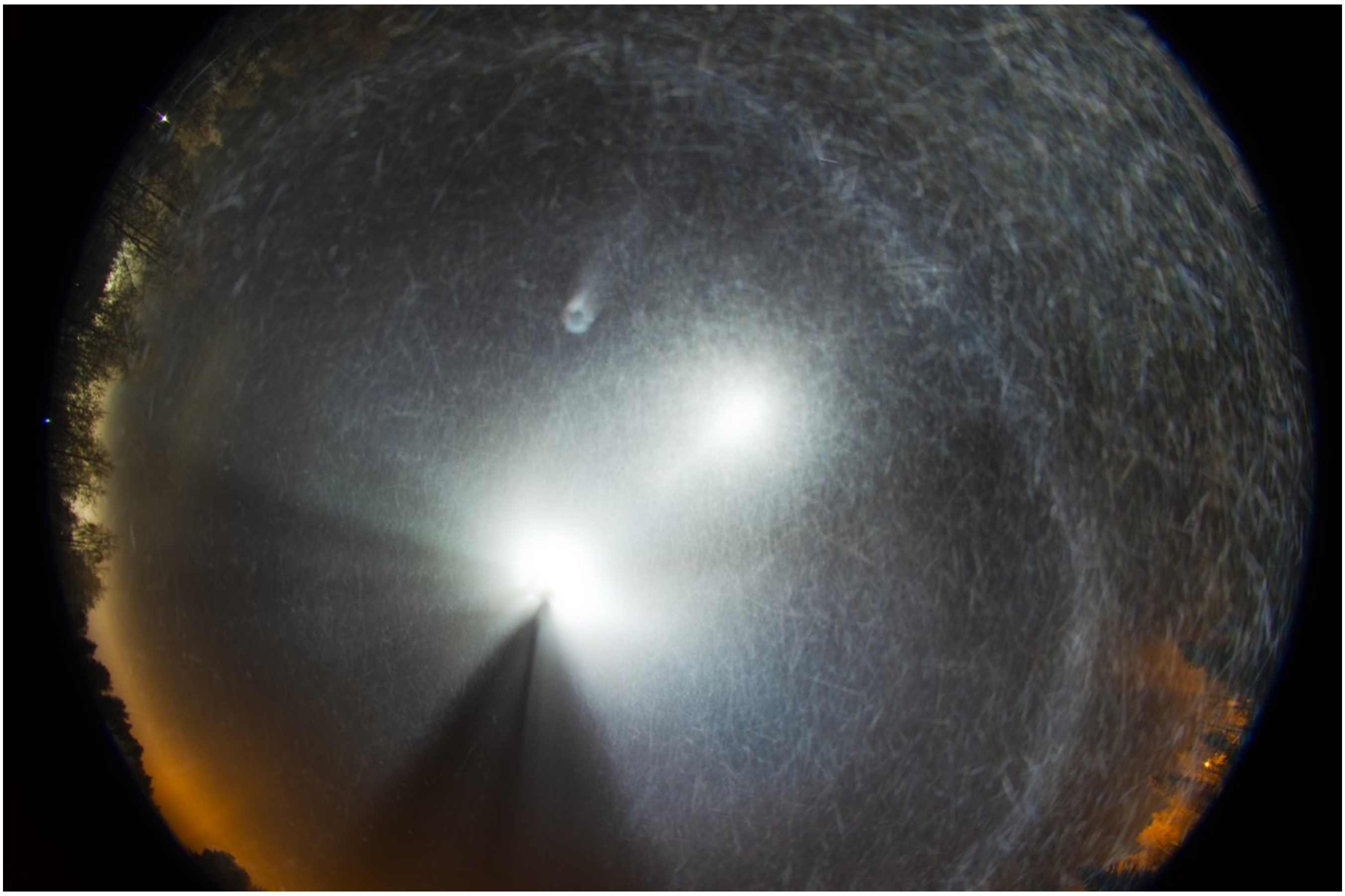
3/9 - 2024-12-01 13:03 UTC

Two more images (ave and max stack) with the lamp moved again, this time closer to the camera. I should next time go the whole hog and have the lamp directly underneath the camera. I need to look at my tripods, maybe one of them allows to place the central bar horizontally to make the operation easier.

The third image has again the lamp moved, this time further away to make the elevation moderate enough to allow for the counterparhelia. After this the ice fog vanished and it was time head back to apartment.





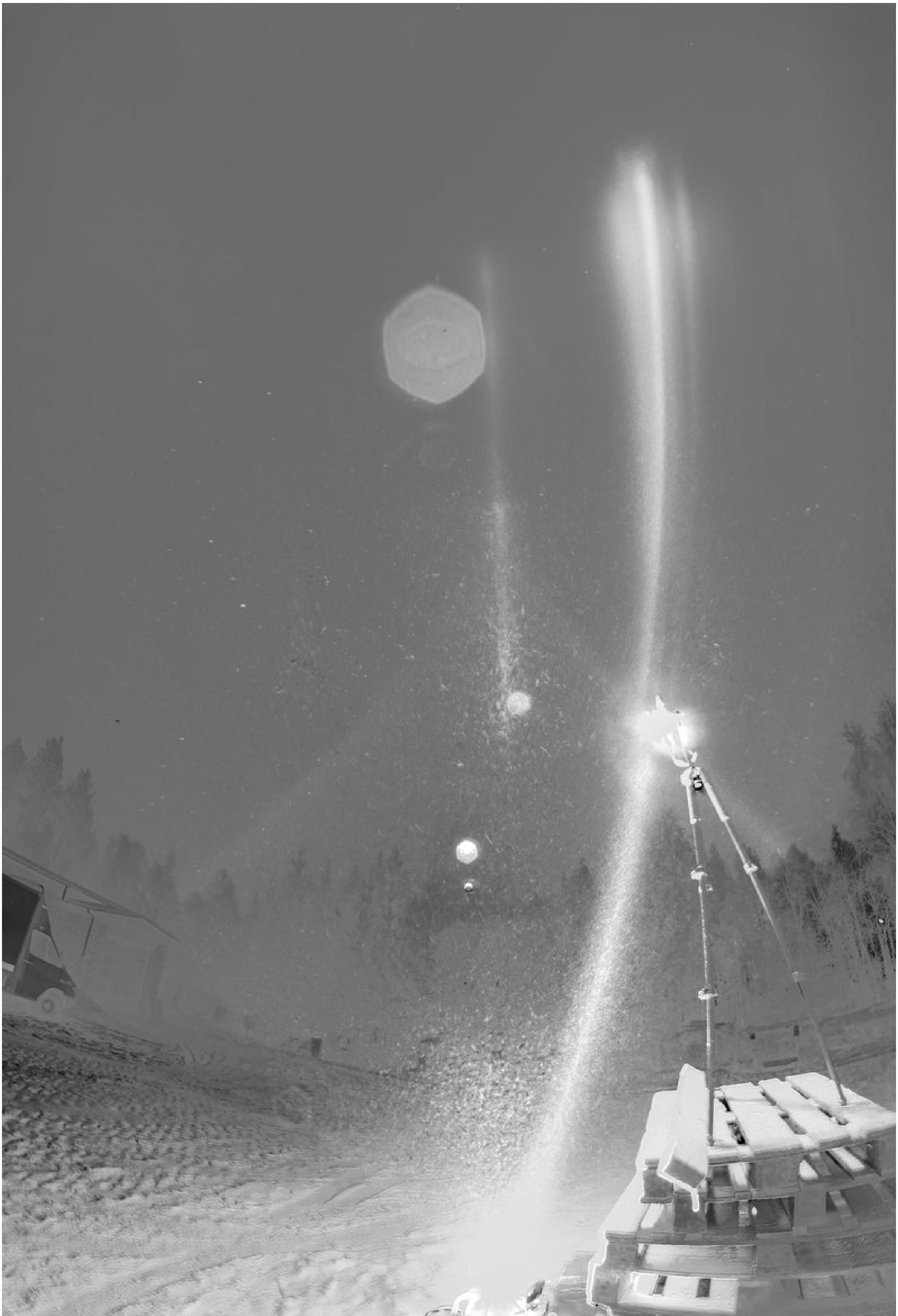


4/9 - 2024-12-02 04:30 UTC

Photographing from the side gives non-classical instances of halos in the spill of the beam. The first two images are 3 frame stacks, the third a single frame. I wonder what the features pointed by arrows are? The middle arrow feature I actually saw with my eyes which prompted to take these photos. Need to consult Streetlight Halos.

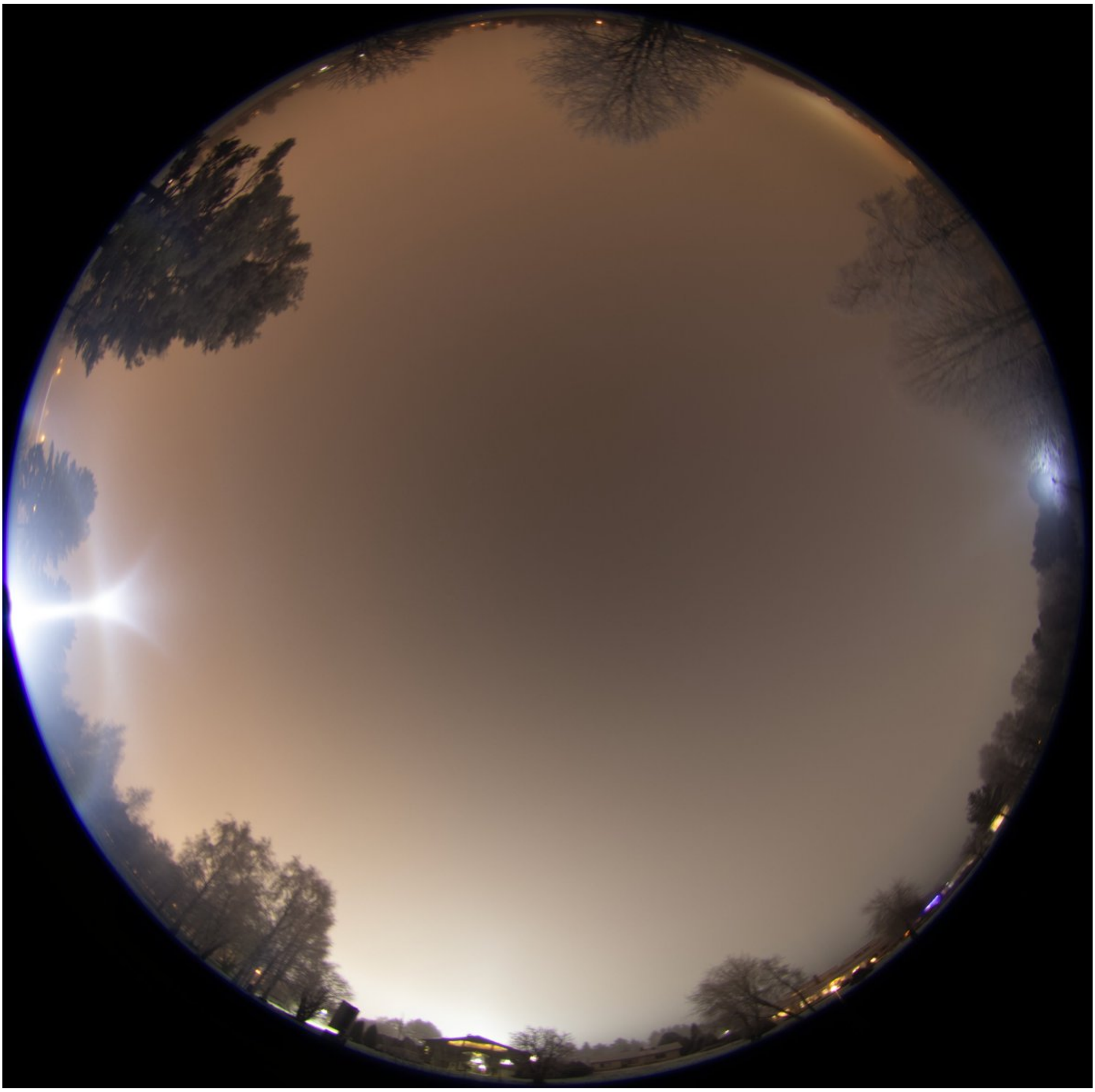
Camera was at about 160 cm height. In the last photo the camera was on the ground, the arrow marks a feature that maybe is the same as the one marked the middle arrow in the other photo.

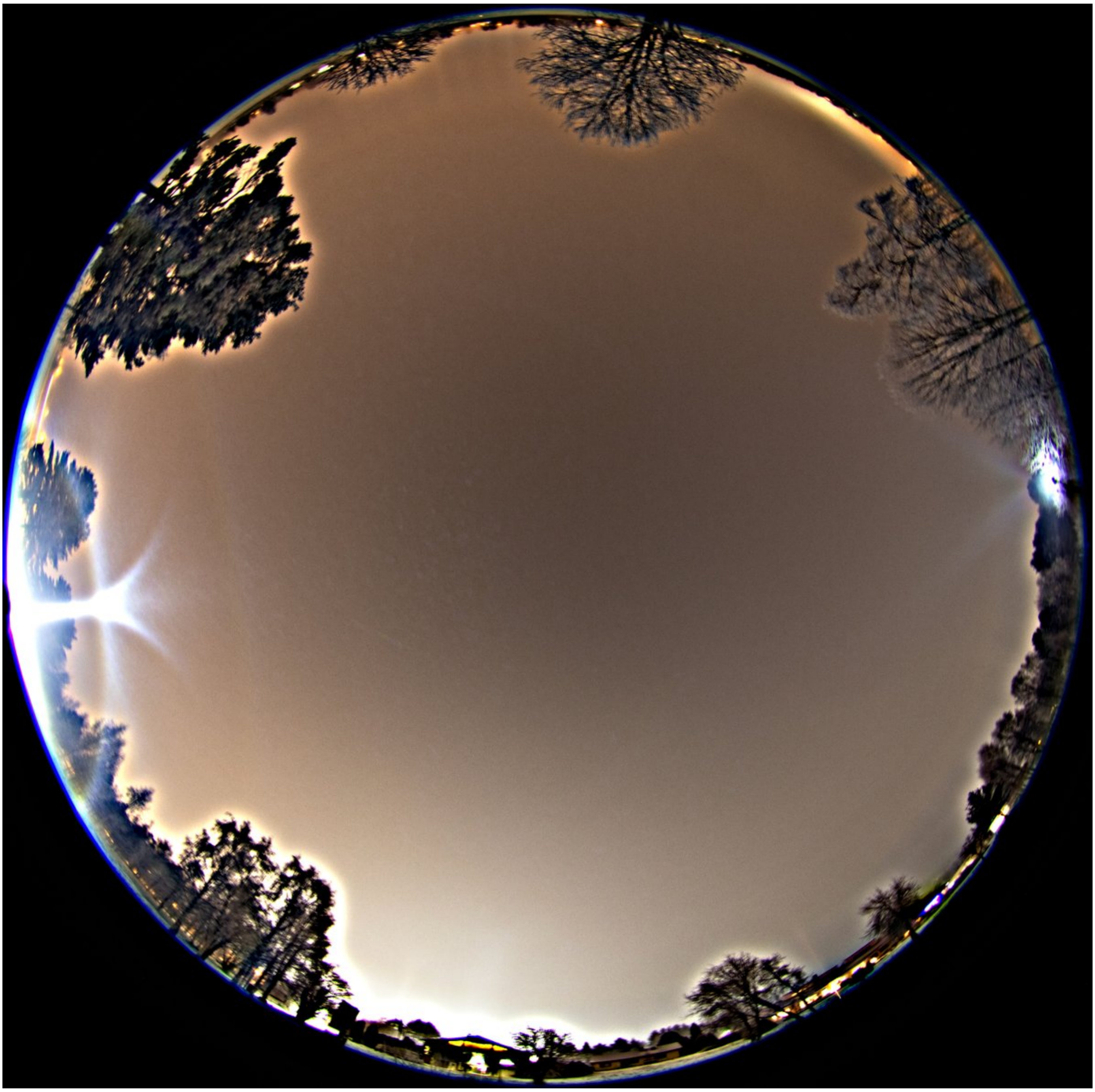
Anyway, if you are chasing new halos which classical instances require a very low lamp, maybe all that hassle of getting the camera high above the lamp is unnecessary if you could bag their spatial instances from the side like here. I am already thinking of bright flood led torches.

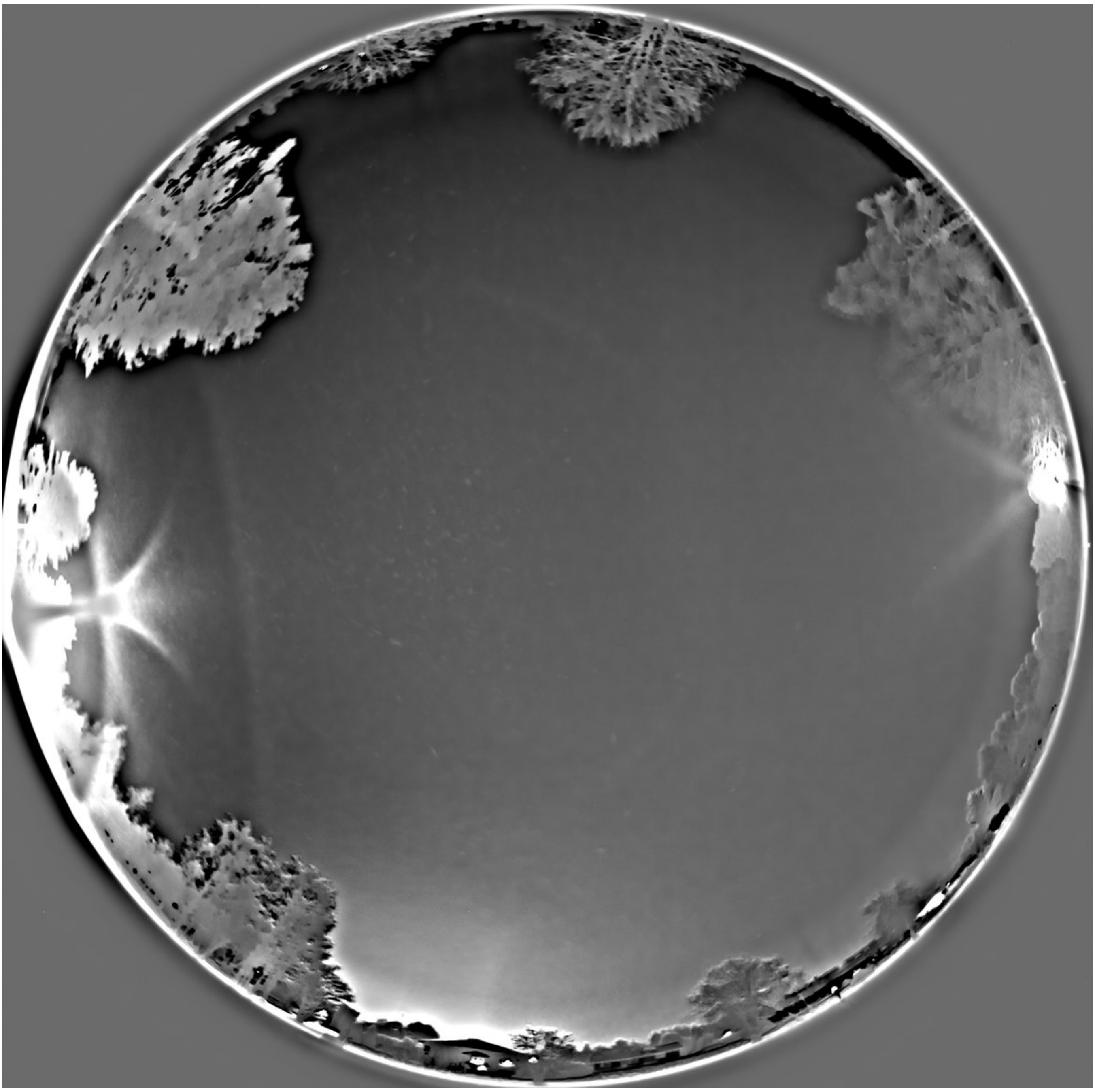


5/9 - 2024-12-10 06:38 UTC

More stuff from the 28/29 November 2024 Rovaniemi night. These were before and in a different place than the very low lamp views in the above posts. Much light pollution due to the fog / low Stratus conditions. We see an enhanced 'Wegener' next to tangent arc. It is quite common feature. Past displays have taught us it does no necessarily curve horizontally like true Alfie but slopes downwards and tends to be associated with the more crappy phases of the display. I think got it nicely photographed in an earlier display this winter but the whereabouts of the memory card containing those shots is currently unknown.

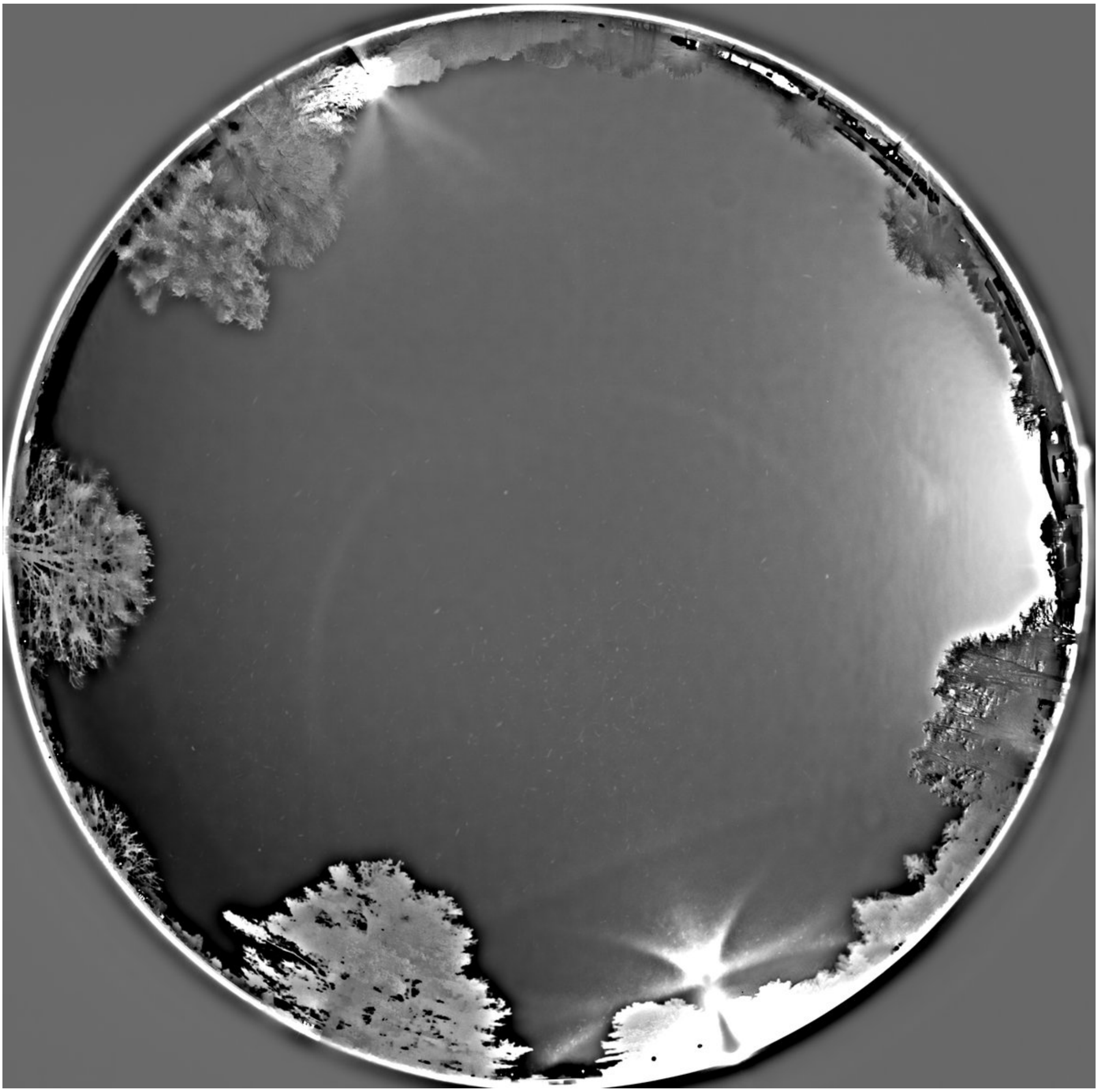


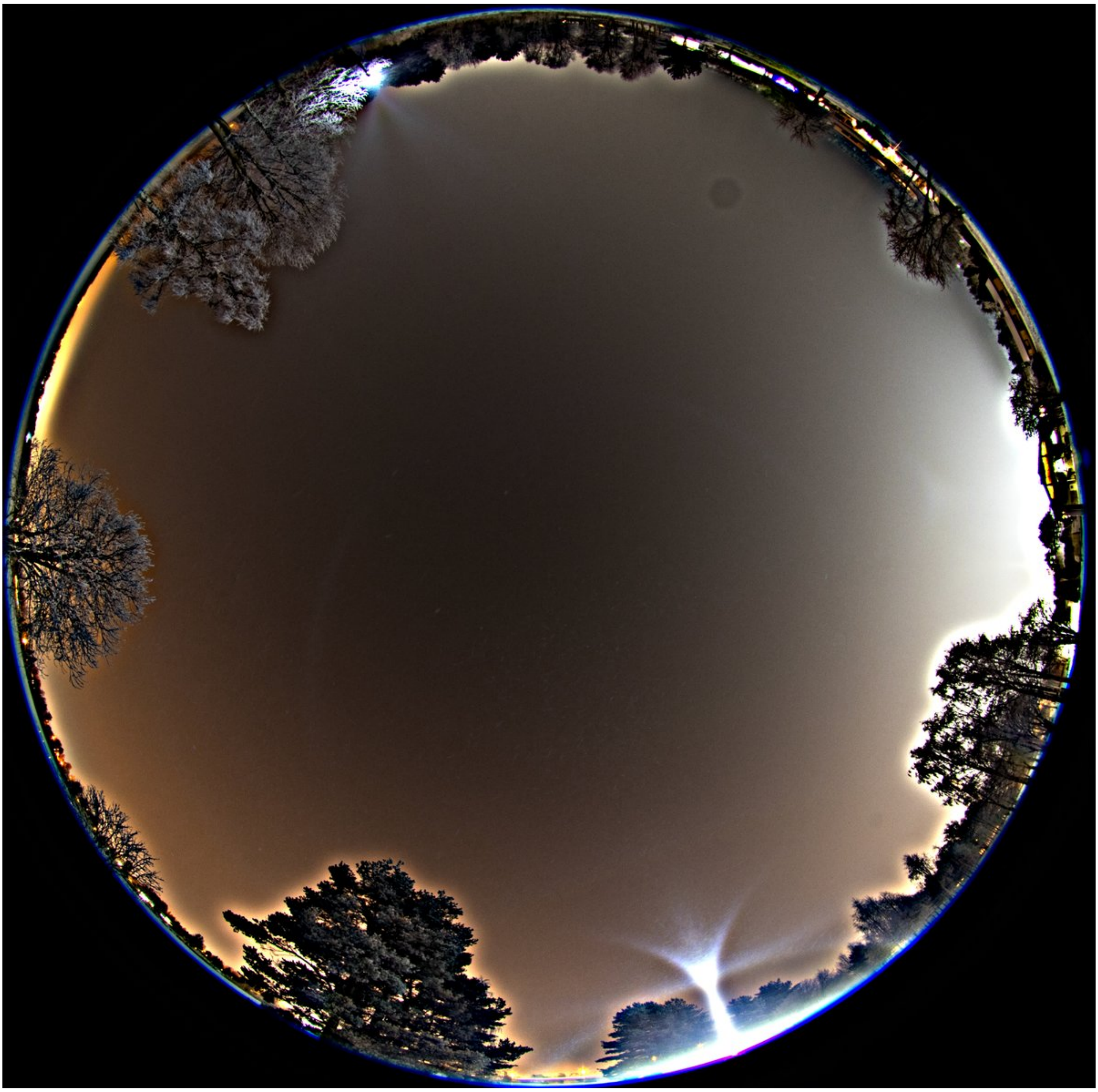




6/9 - 2024-12-10 06:38 UTC

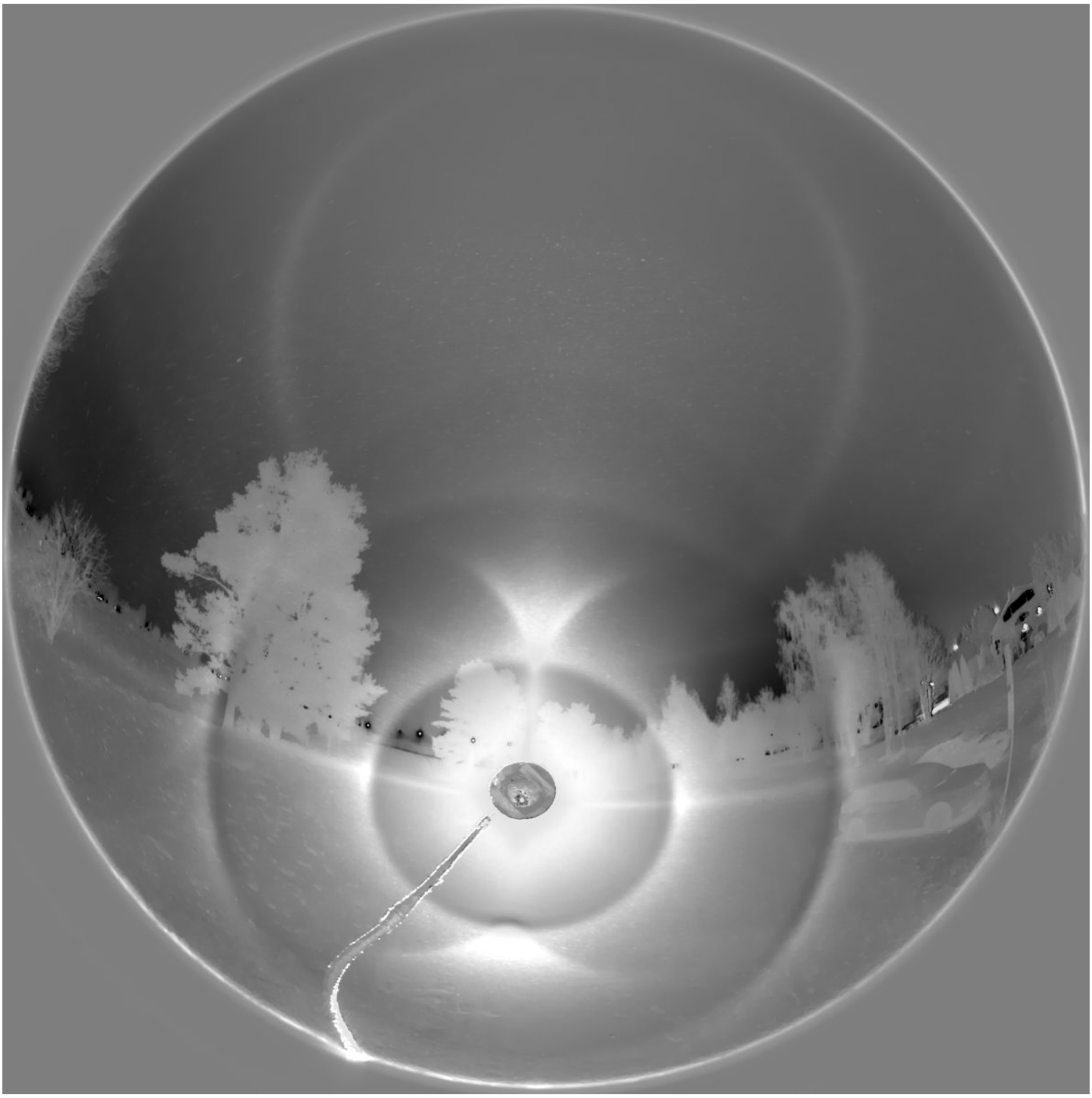
More of the same as above. 28/29 November 2024 Rovaniemi.





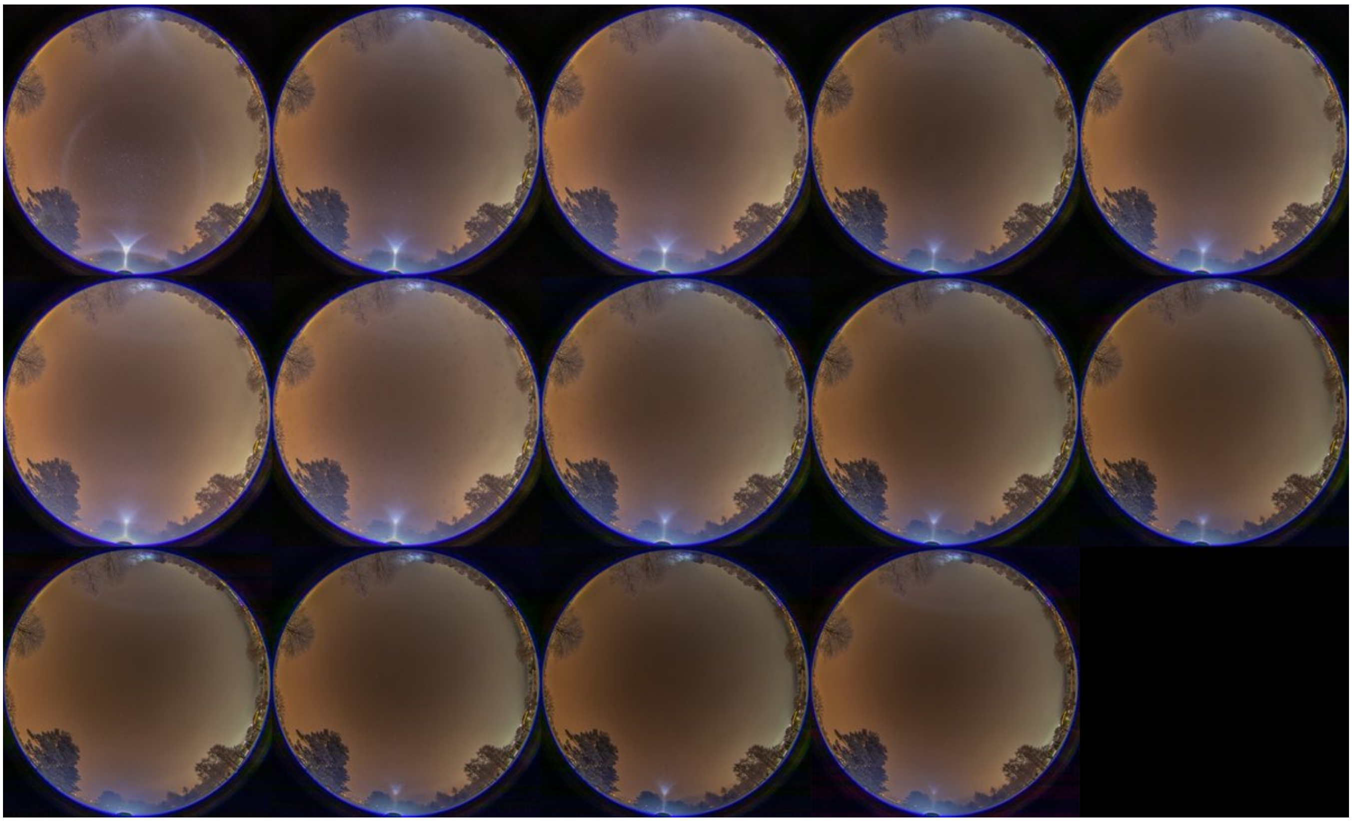
7/9 - 2024-12-10 06:38 UTC

28/29 November 2024 Rovaniemi. For some 5 minutes or so the display amped up. Nice counterhelic arc here. No Parry orientation at all. Notice how the anomalous Alfie is gone in this high quality stage. And while parhelia is visible there is no counterparhelia (subparhelia). After the lamp side shots I tilted the camera upwards, but managed to get only one ptbefore the action weakened. This third photo here is a bit messed up because exposure was on while tilting the camera.



8/9 - 2024-12-10 06:38 UTC

28/29 November 2024 Rovaniemi. Here are the immediate shots following the third image in the above post. You see a fogbow making an appearance, always at the cost of halos.



9/9 - 2024-12-10 06:38 UTC

28/29 November 2024 Rovaniemi. And this was the end of it in this location. It is pretty much the same script every time when ice fog is about to recede from whatever location you are: there is a short stage of big crystals giving a display of high clarity but in a relatively sparse swarm. This is max stack of just a few frames.

I managed some crystals from the earlier stages. Not good photos, I'll see if I add them here later.